Combinatorial Algorithms (Math 482), 2026 Spring Problem Set 11

Due by Wednesday, April 15, 2026.

1. **Updating maximum flow.** We are given a capacitated network, $G = (V, E)$ with source s, sink t, and integer capacities $c(e)$. We are also given a maximum flow f (with flow values $f(e)$ on each edge e).

   (a) Suppose that the capacity of a single edge $(u, v) \in E$ increases by 1. Give an $O(m + n)$-time algorithm to update the maximum flow.

   (b) Suppose that the capacity of a single edge $(u, v) \in E$ decreases by 1. Give an $O(m + n)$-time algorithm to update the maximum flow.

2. **Minimum cut in undirected graphs.** For a capacitated network with a source s and sink t, we defined an st-cut as a partition of the vertices into two parts, A and B, such that $s \in A$ and $t \in B$. In general, we can define a *cut* in an *undirected* graph $G = (V, E)$ (and no source or sink) as a partition of the vertices $V = A \cup B$, and the *size* of the cut is the total number of edges between A and B, that is , $\#\{ab \in E : a \in A, b \in B\}$.

   (a) Show that one can find a minimum cut in a given undirected graph $G = (V, E)$, using a reduction to the maximum flow problem: Running the max-flow algorithm $\binom{n}{2}$ times for all pairs of vertices, setting the two vertices to be a source and a sink.

   (b) Design algorithm that finds a minimum cut of an undirected graph $G = (V, E)$ by running $O(n)$ max-flow problems. What is the running time of your algorithm in terms of m and n?

3. **Maximal versus Maximum Matchings.**

   (a) Show that if M_1 is a maximal matching and M_2 is a maximum matching in a bipartite graph $G = (X \cup Y, E)$, then $|M_1| \leq |M_2| \leq 2 \cdot |M_1|$.

   (b) For every pair of positive integers, $0 < k \leq \ell \leq 2k$, construct a bipartite graph $G = (X \cup Y, E)$ such that there is a maximal matching of size k, but the maximum matching has size ℓ.

   (c) Construct a bipartite graph $G = (X \cup Y, E)$ on 10 vertices and at least 10 edges such that *every* maximal matching is necessarily a maximum matching.

4. **Symmetric difference of two matchings.** Let $G = (V, E)$ be an undirected graph, and let M_1 and M_2 be two matchings.

   (a) Prove that the *symmetric difference* $M_1 \oplus M_2$ is a disjoint union of paths and cycles. (This claim was stated in class without proof.)

   (b) Show that every cycle in $M_1 \oplus M_2$ is an even cycle (that is, it has an even number of edges).

   (c) Prove that if $|M_1| \neq |M_2|$, then $M_1 \oplus M_2$ contains a path with an odd number of edges.